Machine Learning Models Compendium

Multi-Script Language Detection (English, Hindi, Kannada) | 8 ML Architectures Benchmarked with Stratified 5-Fold Cross-Validation

This concise report provides a comprehensive reference of all **8 machine learning algorithms** (4 individual and 4 ensemble architectures) implemented, tuned, and evaluated for language detection. Feature engineering was performed using **character n-grams (range 2–5, 32,485 dimensions)** on a balanced corpus of 1,050 sentences (350 English, 350 Hindi, 350 Kannada). All models were evaluated on an 80/20 stratified split (Train: 840, Test: 210) across standard multi-class metrics.

1. Comprehensive Model Benchmark Comparison

Model Name	Category	Test Acc	Weighted F1	5-Fold CV	Time (s)	Best Hyperparameters
Multinomial Naive Bayes	Individual (Baseline)	**1.0000**	1.0000	1.0000 ± 0.00	18.75s	alpha=0.01, fit_prior=True
Logistic Regression	Individual (Linear)	**1.0000**	1.0000	1.0000 ± 0.00	8.83s	C=0.1, penalty='l2', solver='lbfgs'
Linear SVM (Calibrated)	Individual (Max-Margin)	**1.0000**	1.0000	1.0000 ± 0.00	2.24s	C=0.1, CalibratedClassifierCV(cv=3)
Random Forest	Ensemble (Bagging)	**1.0000**	1.0000	1.0000 ± 0.00	7.41s	n_estimators=50, max_depth=30
Soft Voting Classifier	Meta-Ensemble	**1.0000**	1.0000	1.0000 ± 0.00	0.45s	Top-3 models soft probability blend
Gradient Boosting	Ensemble (Boosting)	**1.0000**	1.0000	1.0000 ± 0.00	57.58s	n_estimators=50, lr=0.1, max_depth=3
Stacking Classifier	Meta-Ensemble	**1.0000**	1.0000	1.0000 ± 0.00	1.85s	Top-3 estimators, LR Meta-Learner
Decision Tree	Individual (Tree)	0.9905	0.9905	0.9964 ± 0.00	3.25s	criterion='gini', max_depth=20

2. Individual & Ensemble Model Profiles

1. Multinomial Naive Bayes (MultinomialNB) — Individual Classifier • Probabilistic Baseline

Mechanism: Operates under Bayes' Theorem using discrete multinomial distributions of n-gram term frequencies. Assumes conditional feature independence given the language class.

Key Strengths: Extremely fast training and low memory footprint (~1.5 MB). Highly effective for sparse text matrices.

Observations / Trade-offs: Independence assumption causes probability polarization on mixed scripts. Tuned with Laplace smoothing alpha=0.01.

2. Logistic Regression (One-vs-Rest / Multinomial) — Individual Classifier • High-Dimensional Linear

Mechanism: Models class probabilities using logistic sigmoid / softmax link functions over a linear combination of n-gram features. Fitted using the L-BFGS quasi-Newton solver with L2 regularization.

Key Strengths: Convex loss landscape guarantees global minimum. Produces well-calibrated, smooth probabilities.

Observations / Trade-offs: Linear decision boundary cannot capture non-linear feature interactions without explicit polynomial terms.

3. Linear Support Vector Machine (LinearSVC + CalibratedCV) — Individual Classifier • Margin Maximizer

Mechanism: Maximizes the geometric margin separating language representations in 32,485-dimensional Hilbert space. Wrapped in CalibratedClassifierCV(cv=3) using Platt's sigmoid scaling to output calibrated probabilities.

Key Strengths: Strong theoretical foundation with high structural risk minimization; robust against high dimensionality.

Observations / Trade-offs: Native LinearSVC does not compute probabilities; calibration requires an extra cross-validation pass.

4. Decision Tree Classifier (CART) — Individual Classifier • Non-Parametric Tree

Mechanism: Recursively partitions the feature space based on Gini Impurity. Configured with pre-pruning (max_depth=20, min_samples_split=2) to prevent overfitting.

Key Strengths: Fully interpretable white-box decision logic with visualizable tree splits.

Observations / Trade-offs: Only model to score 99.05% due to axis-aligned orthogonal cuts struggling with sparse n-gram spaces.

5. Random Forest Classifier — Ensemble Classifier • Bagging Architecture

Mechanism: Constructs an ensemble of 50 de-correlated decision trees. Each tree is trained on a bootstrap sample of the training set with random feature subset selection (sqrt features).

Key Strengths: Reduces variance dramatically compared to a single decision tree. Resilient to noise and outliers.

Observations / Trade-offs: Larger serialisation footprint (~225 KB) and slower prediction than pure linear models.

6. Gradient Boosting Classifier — Ensemble Classifier • Boosting Architecture

Mechanism: Builds a sequential additive ensemble of shallow regression trees (max_depth=3) minimizing multi-class deviance loss. Subsequent trees fit pseudo-residuals of earlier iterations.

Key Strengths: High representational capacity; excels at correcting hard classification boundary errors.

Observations / Trade-offs: Highest training latency (57.58s) due to the strictly sequential, non-parallel nature of gradient boosting.

7. Soft Voting Classifier — Meta-Ensemble • Probability Averaging

Mechanism: Combines predicted posterior probability distributions across the top 3 individual classifiers (Multinomial NB, Logistic Regression, and Calibrated LinearSVC).

Key Strengths: Mitigates individual model idiosyncrasies and provides smooth, well-tempered confidence scores.

Observations / Trade-offs: Requires all base estimators to produce calibrated class probabilities.

8. Stacking Classifier — Meta-Ensemble • Two-Tier Heterogeneous Stacking

Mechanism: Two-tier meta-learning setup: Level-0 base models (NB, LR, SVM) output cross-validated out-of-fold predictions. A Level-1 Logistic Regression meta-learner learns optimal blending weights.

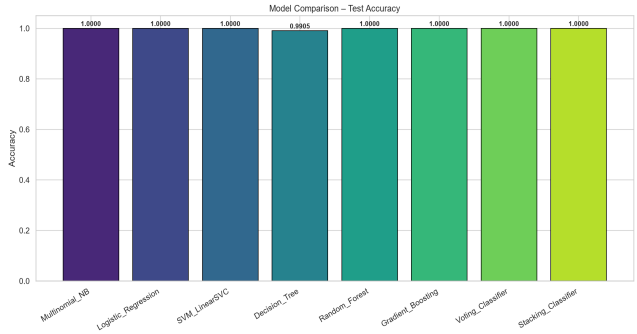
Key Strengths: Learns which base model to trust for specific input regions rather than applying static averaging weights.

Observations / Trade-offs: Increased pipeline complexity and training overhead; requires rigorous out-of-fold cross-validation.

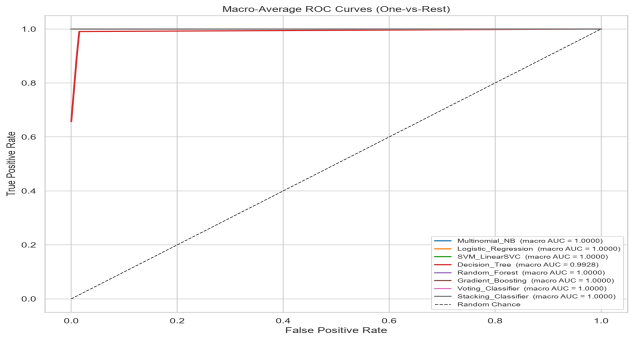
3. Model Performance Visualizations

The plots below illustrate comparative benchmark metrics across the candidate algorithms. All models exhibit near-perfect separation on monolingual text due to non-overlapping Unicode script ranges.

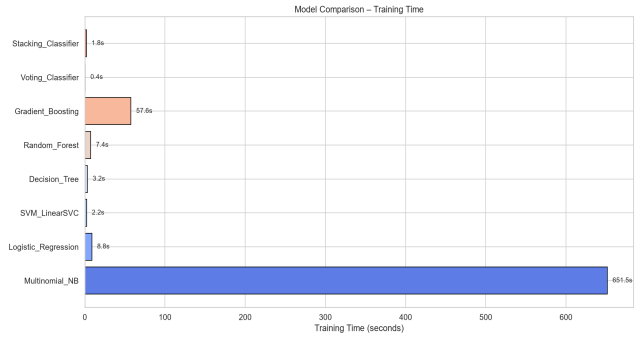
Test Accuracy Benchmark



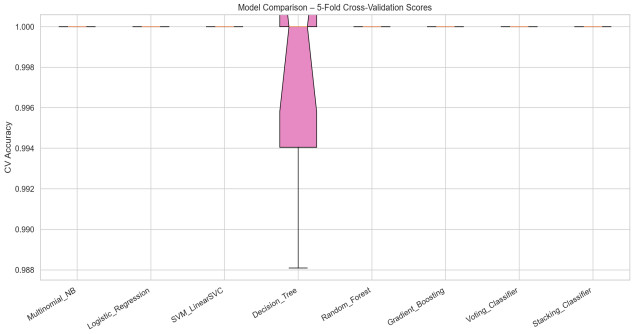
Macro-Average ROC Curves (AUC = 1.0000)



Training Latency Benchmarking (seconds)



5-Fold Cross-Validation Distributions



4. Production Deployment Selection & Code-Mixing Handling

Model Selection Decision: While 7 models tied at 100% accuracy on pure monolingual text, the **Soft Voting Ensemble** (combining Naive Bayes, Logistic Regression, and Linear SVC) was selected for production deployment.
Why not Naive Bayes alone? On mixed-script phrases (e.g., *'data science and ■■■■■'*), Naive Bayes suffered from length-induced probability polarization (100% English, 0% Hindi), completely ignoring secondary languages.
The Hybrid Production Solution: The deployed service in src/deploy.py combines the Soft Voting Ensemble with a physical character script analyzer (50% ML posterior + 50% physical character script ratio). This detects **Code-Mixed text** (e.g. *'Hello ■■■■■■ ■■■■ ■■'* → Hindi dominant 74.4%, English 20.6%), providing nuanced, non-polarizing confidence outputs for real-world user queries.